

LECTURE 6: INDEPENDENCE

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

Please email me with any observed typos to elre@princeton.edu, thanks in advance!

E. Rebrova

When we talk about individual random variables, we are typically just interested in their laws (e.g., the Gaussian distribution, exponential distribution, Bernoulli distribution, etc.). The law $\mathcal{P}_X$, cumulative distribution function F_X , probability density function f_X , etc. define the expectation of a random variable X and its moments. It does not really matter what the underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is, and which mapping X represents the law.

However, as soon as we start talking about collections of random variables, it matters how their values depend on each other, so we have to look back at the mapping $X(\omega)$.

DEFINITIONS

Independence can be quantified on three levels:

- independence of events;
- independence of σ -algebras;
- independence of random variables.

Definition 6.1 (Independence of events). Two events $A, B \in \mathcal{F}$ are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Example. It is not always very intuitive which events are independent. For example, suppose that we model an experiment where we throw two fair six-sided dice. Consider three events:

$$A_1 = \{ \text{sum of two numbers is 6} \},$$

$$A_2 = \{ \text{sum of two numbers is 7} \},$$

$$A_3 = \{ \text{first die is 4} \}.$$

Which pairs of these events are independent? What about the others?

Definition 6.2 (Independence of σ -algebras). Two σ -algebras $\mathcal{F}_1, \mathcal{F}_2 \subseteq \mathcal{F}$ are independent if for any $A \in \mathcal{F}_1$ and $B \in \mathcal{F}_2$, the events A and B are independent (i.e., $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$).

Definition 6.3 (Independence of random variables). Two random variables $X_1, X_2 : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$ are independent if their generated σ -algebras $\sigma(X_1), \sigma(X_2) \subseteq \mathcal{F}$ are independent. The definition is naturally extended if we want to say that two collections of random variables are independent.

Note that the independence of events and random variables is well-aligned. In particular, two events A, B are independent if and only if the corresponding indicator random variables $\mathbb{1}_A, \mathbb{1}_B$ are independent.

To see this, suppose that A, B are independent (denoted $A \perp\!\!\!\perp B$). Note that if $A \perp\!\!\!\perp B$, then $A \perp\!\!\!\perp B^c$, $A^c \perp\!\!\!\perp B$, and $A^c \perp\!\!\!\perp B^c$. Also, $\emptyset$ and Ω are independent with any other set (check these claims!). So, the σ -algebras $\sigma(\mathbb{1}_A) = \{\emptyset, A, A^c, \Omega\}$ and $\sigma(\mathbb{1}_B) = \{\emptyset, B, B^c, \Omega\}$ are independent. Hence, the random variables $\mathbb{1}_A$ and $\mathbb{1}_B$ are also independent.

Definition 6.4 (Mutual independence). A collection $\{A_\alpha\}_{\alpha \in I}$ of events in $\mathcal{F}$ (where the indices I are possibly infinite or uncountable) is mutually independent if any finite subset of events $A_1, A_2, \dots, A_\ell$ satisfies

$$\mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_\ell) = \prod_{i=1}^{\ell} \mathbb{P}(A_i).$$

A collection $\{\mathcal{F}_\alpha\}_{\alpha \in I}$ of σ -algebras with $\mathcal{F}_\alpha \subseteq \mathcal{F}$, $\alpha \in I$, is mutually independent if for any finite subset of distinct indices $\alpha_1, \alpha_2, \dots, \alpha_\ell$, we have

$$\mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_\ell) = \prod_{i=1}^{\ell} \mathbb{P}(A_i) \text{ for any } A_i \in \mathcal{F}_{\alpha_i}, i = 1, 2, \dots, \ell.$$

A collection of random variables is mutually independent if the respective generated σ -algebras are mutually independent.

JOINT CDF AND JOINT LAW OF MUTUALLY INDEPENDENT RANDOM VARIABLES

Theorem 6.5. Random variables $X_1, \dots, X_m$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ are mutually independent if and only if their **joint CDF** is the product of their individual CDFs: namely,

$$\mathbb{P}(X_1 \leq t_1, \dots, X_m \leq t_m) = \prod_{i=1}^m \mathbb{P}(X_i \leq t_i) \text{ for any } t_1, \dots, t_m \in \mathbb{R}.$$

Proof. One direction is easy (if $X_1, \dots, X_m$ are mutually independent, then their joint CDF factorizes), so we will focus on proving the converse. Let $\mathcal{A}_i = \{X_i^{-1}((-\infty, b]) : b \in \mathbb{R}\}$ for $i = 1, \dots, m$. These form a collection of π -systems (check this!) that generates the collection $\sigma(X_1), \dots, \sigma(X_m)$ of σ -algebras associated with the random variables $X_1, \dots, X_m$. We can then apply the following technical Theorem 6.6 to conclude that $\sigma(X_1), \dots, \sigma(X_m)$ are also mutually independent. $\square$

Theorem 6.6. Let $\{\mathcal{A}_i\}_{i \in I}$ be a collection of π -systems,¹ and $\{\mathcal{F}_i\}_{i \in I}$ with $\mathcal{F}_i = \sigma(\mathcal{A}_i)$ be a collection of the corresponding generated σ -algebras. If $\{\mathcal{A}_i\}_{i \in I}$ is mutually independent, then $\{\mathcal{F}_i\}_{i \in I}$ is mutually independent as well.

Proof. It is enough to show that if a finite subset $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$ are mutually independent, then the corresponding generated σ -algebras $\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_n$ are also mutually independent (think this through!).

To show this claim, let $A_2, \dots, A_n$ be any events with $A_i \in \mathcal{F}_i$, $i = 2, \dots, n$, and let $H = A_2 \cap \dots \cap A_n$. Consider the two measures $\mathbb{P}_1, \mathbb{P}_2$ on $(\Omega, \sigma(\mathcal{A}_1))$ defined by

$$\mathbb{P}_1(G) = \mathbb{P}(G \cap H) \quad \text{and} \quad \mathbb{P}_2(G) = \mathbb{P}(G)\mathbb{P}(H) \quad \forall G \in \sigma(\mathcal{A}_1).$$

¹Recall from Lecture 2 that a π -system is a collection of subsets of Ω that is closed under finite intersections.

It is easy to see that $\mathbb{P}_1$ and $\mathbb{P}_2$ coincide on $G = \Omega$ as well as on the events $G \in \mathcal{A}_1$ in the π -system $\mathcal{A}_1$ by assumption. So, by the corollary of Dynkin's π - λ theorem (from Lecture 2), they must coincide on the whole σ -algebra $\sigma(\mathcal{A}_1) = \mathcal{F}_1$. By continuing with the same argument (or more formally, by using induction), this shows that $\mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_n) = \mathbb{P}(A_1)\mathbb{P}(A_2) \dots \mathbb{P}(A_n)$ for all $A_i \in \sigma(\mathcal{F}_i)$, $i = 1, 2, \dots, n$. $\square$

Definition 6.7 (Joint law). The **joint law** of a collection of random variables $X_1, \dots, X_n$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is the probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ defined by

$$\mathcal{P}_{\bar{X}}(A) := \mathbb{P}(\{\omega \in \Omega : (X_1(\omega), X_2(\omega), \dots, X_n(\omega)) \in A\})$$

for all $A \in \mathcal{F}$. Here, $\bar{X}$ denotes the vector $(X_1, \dots, X_n)$.

We will show in Theorem 6.11 that the joint law of mutually independent random variables satisfies the following natural property: the probability of rectangular sets of the form $A_1 \times \dots \times A_n$ is equal to the product of probabilities of individual sets A_i . Now, we need to extend this property to all measurable sets beyond the rectangular sets. Note that these sets belong to the *product space* $\Omega := \Omega_1 \times \Omega_2 \times \dots \times \Omega_n$.

Definition 6.8 (Product space). Suppose that $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i = 1, 2, \dots, n$, are measure spaces, and let $\Omega := \Omega_1 \times \Omega_2 \times \dots \times \Omega_n$ be the Cartesian product of the sets Ω_i (i.e., the space of all n -tuples). Define the **product σ -algebra**

$$\mathcal{F} = \mathcal{F}_1 \times \mathcal{F}_2 \times \dots \times \mathcal{F}_n := \sigma(\{A_1 \times A_2 \times \dots \times A_n : A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2, \dots, A_n \in \mathcal{F}_n\})$$

to be the σ -algebra generated by all “rectangles” of the form $A_1 \times \dots \times A_n$ with $A_i \in \mathcal{F}_i$. Finally, define the **product measure** μ on the measurable space $(\Omega, \mathcal{F})$ by

$$\mu(A_1 \times \dots \times A_n) := \prod_{i=1}^n \mu_i(A_i) \text{ for each } A_i \in \mathcal{F}_i.$$

Lemma 6.9. *The product measure μ defined above exists and is unique.*

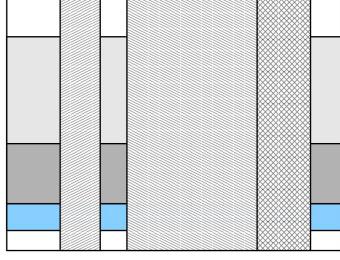
Proof. Our goal is to invoke Carathéodory's extension theorem (from Lecture 2):

- (1) Consider the algebra of finite disjoint unions of rectangles (which generates the product σ -algebra);
- (2) Check that μ is a measure (in particular, σ -additive) on this algebra – this is the main technical step.
- (3) Use Carathéodory's extension theorem to conclude – done!

We will focus on checking σ -additivity of μ : for simplicity, we will consider the product space $\Omega = \Omega_1 \times \Omega_2$ with product σ -algebra $\mathcal{F} = \mathcal{F}_1 \times \mathcal{F}_2$. Let

$$\mathcal{A} = \left\{ \bigsqcup_{i=1}^n A_{1i} \times A_{2i} : A_{1i} \in \mathcal{F}_1, A_{2i} \in \mathcal{F}_2, n \geq 0 \right\}$$

be the algebra of finite disjoint unions of rectangles of the form $A_{1i} \times A_{2i}$ with $A_{1i} \in \mathcal{F}_1$, $A_{2i} \in \mathcal{F}_2$ (check that this is indeed an algebra!). Clearly, $\mathcal{F} = \sigma(\mathcal{A})$ by definition of the product σ -algebra. Suppose that we have a rectangle $A_1 \times A_2 = \bigsqcup_j (A_{1j} \times A_{2j})$ that can be written as a countable disjoint union of rectangles with $A_{1j} \in \mathcal{F}_1$, $A_{2j} \in \mathcal{F}_2$.



Note that the individual sets A_{1j} are not necessarily disjoint. However, for any $(x, y) \in \Omega_1 \times \Omega_2$, we can write

$$\mathbb{1}_{A_1}(x)\mathbb{1}_{A_2}(y) = \sum_j \mathbb{1}_{A_{1j}}(x)\mathbb{1}_{A_{2j}}(y).$$

By integrating with respect to y and the measure μ_2 , keeping x fixed, and using linearity of the integral and the monotone convergence theorem, we obtain

$$\mathbb{1}_{A_1}(x)\mu_2(A_2) = \sum_j \mathbb{1}_{A_{1j}}(x)\mu_2(A_{2j})$$

Similarly, by integrating with respect to x , we get

$$\mu_1(A_1)\mu_2(A_2) = \sum_j \mu_1(A_{1j})\mu_2(A_{2j}).$$

Note that by definition of μ , $\mu(A_{1j} \times A_{2j}) = \mu_1(A_{1j})\mu_2(A_{2j})$ for all j .

Therefore, if $A \in \mathcal{A}$ is a countable disjoint union of subsets in $\mathcal{A}$, we may write $A = \sqcup_\ell (A_{1\ell} \times A_{2\ell})$ or $A = \sqcup_{i=1}^n (A_{1i} \times A_{2i})$, where each $A_{1i} \times A_{2i} = \sqcup_j (A_{1ij} \times A_{2ij})$. Hence, from what we have shown, $\mu(A) = \sum_{i=1}^n \sum_j \mu(A_{1ij} \times A_{2ij})$. The proof is completed by observing that $\sum_\ell \mu(A_{1\ell} \times A_{2\ell}) = \sum_\ell \sum_{i=1}^n \sum_j \mu((A_{1\ell} \times A_{2\ell}) \cap (A_{1ij} \times A_{2ij})) = \sum_{i=1}^n \sum_j \mu(A_{1ij} \times A_{2ij})$.² $\square$

Remark 6.10. The construction above proves the existence and uniqueness of the Lebesgue measure on $\mathbb{R}^n$. (How?)

Theorem 6.11. A collection of RVs $X_1, \dots, X_n$ on $(\Omega, \mathcal{F}, \mathbb{P})$ is mutually independent if and only if their joint law $\mathcal{P}_{\bar{X}}$ satisfies $\mathcal{P}_{\bar{X}} = \mathcal{P}_{X_1} \times \mathcal{P}_{X_2} \times \dots \times \mathcal{P}_{X_n}$.

Proof. Suppose that $X_1, \dots, X_n$ are mutually independent. Then for any rectangle $A_1 \times \dots \times A_n \in \mathcal{F}$, we have

$$\begin{aligned} \mathcal{P}_{\bar{X}}(A_1 \times \dots \times A_n) &= \mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \prod_i \mathcal{P}(X_i \in A_i) = \prod_i \mathcal{P}_{X_i}(A_i) \\ &= (\mathcal{P}_{X_1} \times \mathcal{P}_{X_2} \times \dots \times \mathcal{P}_{X_n})(A_1 \times A_2 \times \dots \times A_n). \end{aligned}$$

Since the collection of rectangles of the form $A_1 \times \dots \times A_n$ forms a π -system that generates $\mathcal{B}(\mathbb{R}^n)$ (check this!), the measures (laws) $\mathcal{P}_{\bar{X}}$ and $\mathcal{P}_{X_1} \times \mathcal{P}_{X_2} \times \dots \times \mathcal{P}_{X_n}$ must coincide on the whole σ -algebra by the Dynkin π - λ theorem.

The other side—showing that if the joint law is represented by a product, then the random variables are mutually independent—can be checked by a similar line of equalities verifying the CDF independence condition in the proof of Corollary 6.5. $\square$

²Technically, this requires interchanging two infinite sums, which is possible *since the summands are non-negative*. This can be shown using elementary methods from analysis, however it is a particular instance of a more general result known as Fubini-Tonelli's theorem (which we will see later)

Remark 6.12 (Infinite product spaces). This is important for constructing stochastic processes. Suppose that $\{(\Omega_i, \mathcal{F}_i, \mu_i)\}_i$ is a sequence of measure spaces and $\Omega = \Omega_1 \times \Omega_2 \times \dots$ is the Cartesian product of the spaces (i.e., the space of sequences of elements $(\omega_1, \omega_2, \dots)$ with $\omega_i \in \Omega_i$). The product σ -algebra $\mathcal{F}$ is generated by the “generalized rectangles” or *cylinder sets* of the form $A_1 \times A_2 \times \dots$, where $A_i \in \mathcal{F}_i$ and $A_i = \Omega_i$ for all but finitely many indices i). The existence of a product measure on $(\Omega, \mathcal{F})$ is a slightly trickier question; in particular, we require $\mu(\Omega_i) = 1$ for each i so that we do not multiply infinitely many values that are either less or bigger than 1. The key result for constructing such a product measure is the following theorem.

Theorem 6.13 (Kolmogorov extension theorem). Let $\{\mu_n\}_{n \geq 1}$ be a sequence of probability measures on $\mathbb{R}^n$ for each n . Suppose that the sequence satisfies the consistency property: $\mu_{n+1}(B \times \mathbb{R}) = \mu_n(B)$ for all $B \in \mathcal{B}(\mathbb{R}^n)$ and $n \geq 1$. Then there exists a unique probability measure μ on $(\mathbb{R}^{\mathbb{N}}, \mathcal{B}_c)$ such that

$$\mu(B_1 \times \dots \times B_n \times \mathbb{R} \times \mathbb{R} \times \dots) = \mu_n \quad \text{for any } n \text{ and } B_1, \dots, B_n \in \mathcal{B}(\mathbb{R}).$$

Here, $\mathcal{B}_c$ is the σ -algebra on $\mathbb{R}^{\mathbb{N}}$ generated by all finite Cartesian products of $\mathcal{B}(\mathbb{R})$ -measurable sets of the form $B_1 \times B_2 \times \dots \times B_n \times \mathbb{R} \times \mathbb{R} \times \dots$.

INTEGRATION OVER PRODUCT SPACES: FUBINI-TONELLI THEOREM

Next, we aim to show that the expectation of the product of independent random variables is the product of expectations. To this end, we give a more general result about the integration with respect to product measures:

Theorem 6.14 (Fubini-Tonelli theorem). Suppose that we have two measure spaces $(\Omega_1, \mathcal{F}_1, \mu_1)$ and $(\Omega_2, \mathcal{F}_2, \mu_2)$ where μ_1 and μ_2 are σ -finite measures. Let $\mu = \mu_1 \times \mu_2$. If $f : \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}$ is a measurable function that is either (a) integrable or (b) non-negative, then

$$\int_{\Omega_1 \times \Omega_2} f(x, y) \, d\mu(x, y) = \int_{\Omega_1} \int_{\Omega_2} f(x, y) \, d\mu_2(y) \, d\mu_1(x) = \int_{\Omega_2} \int_{\Omega_1} f(x, y) \, d\mu_1(x) \, d\mu_2(y).$$

We omit the proof. On a very high level, the proof starts from proving the statement for indicator functions and then generalizes to a wider class of functions, either with the standard 4-step argument that we have used before, or with a more general machinery known as the *Monotone Class Theorem* that serves a similar goal (we have not seen this before). In the simplest case when $f(x, y) = \mathbb{1}_{A_1 \times A_2}$ (this is even more restrictive than an arbitrary indicator function), note that the statement follows from the definition of the product measure $\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$.

The following provides a simple counterexample showing that the Fubini-Tonelli theorem can fail if its hypotheses are not satisfied:

$\Omega_1 = \Omega_2 = \mathbb{Z}_+$

$\mu_1 = \mu_2$ counting measures

$f(x, y) = \begin{cases} +1 & \text{if } x=y \\ -1 & \text{if } y=x+1 \\ 0 & \text{otherwise} \end{cases}$

Integration = summing up
(see Lec 5 discrete example)

$\sum_{\text{rows}} \sum_{\text{cols}} = 0 + 0 + 0 \dots = 0$

$\sum_{\text{cols}} \sum_{\text{rows}} = 1 + 0 + 0 \dots = 1$

Q: what assumptions do not hold?

We continue with three useful corollaries of Fubini's.

Corollary 6.15 (Expectations of products). *If X and Y are independent random variables, and $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are measurable functions that are either both non-negative, or satisfy $\mathbb{E}|f(X)| < \infty$ and $\mathbb{E}|g(Y)| < \infty$, then*

$$\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\mathbb{E}[g(Y)].$$

Proof sketch. (For more details, see [Dembo, Theorem 1.4.30].) Note that for two independent random variables X and Y on $\mathbb{R}$ and an integrable (or measurable and non-negative) function $h : \mathbb{R}^2 \rightarrow \mathbb{R}$, we have from independence (Theorem 6.11) and the Fubini-Tonelli theorem (Theorem 6.14) that

$$\begin{aligned} \mathbb{E}h(X, Y) &= \int_{\mathbb{R}^2} h(x, y) \, d\mathcal{P}_{(X, Y)}(x, y) = \int_{\mathbb{R}^2} h(x, y) \, d(\mathcal{P}_X(x) \times \mathcal{P}_Y(y)) \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} h(x, y) \, d\mathcal{P}_X(x) \, d\mathcal{P}_Y(y). \end{aligned}$$

We may then take $h(x, y) = f(x)g(y)$, verify that the required hypotheses are satisfied by our assumptions, and break the integral into two. $\square$

Remark 6.16. Some other familiar results for independent random variables:

- **Product of several RVs.** If $X_1, \dots, X_k$ are mutually independent integrable random variables that are all either non-negative or integrable, then $\mathbb{E}[\prod_{i=1}^k X_i] = \prod_{i=1}^k \mathbb{E}X_i$. To show this, it is convenient to first check that if $X_{k,j}$ for $1 \leq k \leq K$ and $1 \leq j \leq J(k)$ are mutually independent and f_k are measurable functions, then $Y_k = f_k(X_{k,1}, \dots, X_{k,J(k)})$ are also mutually independent. Then we apply this fact to X_1 and $X_2 \dots X_s$ for $s = 2, \dots, k$. (See, e.g., [Dembo, Corollary 1.4.33].)
- **Variance of a sum and covariances.** Another useful fact is that for any random variables $X_1, \dots, X_k$ defined on the same probability space, we have

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i,j=1}^n \text{Cov}(X_i, X_j),$$

where $\text{Cov}(X, Y) = \mathbb{E}((X - \mathbb{E}X)(Y - \mathbb{E}Y))$ is the covariance between X and Y .

What does this imply if $X_1, \dots, X_k$ are mutually independent? What is more restrictive—to require independence or zero pairwise correlations? Think of an example to confirm the answer.

Corollary 6.17 (Integration by parts). *Suppose that f is a measurable non-negative function on $\mathbb{R}$ and $F(x) = \int_{-\infty}^x f(y) \, dy$ for any $x \in \mathbb{R}$. Then for any random variable X ,*

$$\mathbb{E}[F(X)] = \int_{\mathbb{R}} f(x) \mathbb{P}(X > x) \, dx.$$

In particular, if X is non-negative, then

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > x) \, dx.$$

Proof. By using the change of variables theorem (from Lecture 5), we can write

$$\mathbb{E}[F(X)] = \int_{\mathbb{R}} F(x) \, d\mathcal{P}_X(x) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(y) \mathbf{1}_{\{x > y\}} \, dy \right) \, d\mathcal{P}_X(x).$$

Note that for fixed y , $\int_{\mathbb{R}} f(y) \mathbb{1}_{\{x > y\}} d\mathcal{P}_X(x) = \mathbb{E}[f(y) \mathbb{1}_{\{X > y\}}] = f(y) \mathbb{P}(X > y)$. Thus, applying the Fubini-Tonelli theorem (Theorem 6.14) shows that

$$\mathbb{E}[F(X)] = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(y) \mathbb{1}_{\{x > y\}} d\mathcal{P}_X(x) \right) dy = \int_{\mathbb{R}} f(y) \mathbb{P}(X > y) dy. \quad \square$$

Corollary 6.18 (Density of the independent sum and convolution). *Suppose that X and Y are independent random variables with probability density functions f_X and f_Y respectively. Then the random variable $Z = X + Y$ has probability density function*

$$f_Z(z) = \int_{\mathbb{R}} f_X(z - y) f_Y(y) dy.$$

Proof. For any $z \in \mathbb{R}$, the CDF of Z can be written

$$F_Z(t) = \mathbb{P}(X + Y \leq t) = \mathbb{E}[\mathbb{1}_{\{X+Y \leq t\}}] = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \mathbb{1}_{\{x+y \leq t\}} d\mathcal{P}_X(x) \right) d\mathcal{P}_Y(y) \quad (1)$$

by the Fubini-Tonelli theorem (Theorem 6.14). For fixed y , the inner integral is equal to

$$\int_{\mathbb{R}} \mathbb{1}_{\{x+y \leq t\}} d\mathcal{P}_X(x) = \int_{\mathbb{R}} \mathbb{1}_{\{x \leq t-y\}} d\mathcal{P}_X(x) = \mathcal{P}_X((-\infty, t-y]) = \int_{-\infty}^{t-y} f_X(x) dx,$$

since X has density f_X . Observe that with $z := x + y$ changing variables $dz = dx$, we have

$$\int_{-\infty}^{t-y} f_X(x) dx = \int_{-\infty}^{t-y} f_X(z - y) dz = \int_{-\infty}^t f_X(z - y) dz.$$

Therefore, returning back to (1) and then by another application of the Fubini-Tonelli theorem, we deduce that

$$\begin{aligned} F_Z(t) &= \int_{\mathbb{R}} \left(\int_{-\infty}^t f_X(z - y) dz \right) d\mathcal{P}_Y(y) \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \mathbb{1}_{\{z \leq t\}} f_X(z - y) dz \right) d\mathcal{P}_Y(y) \\ &= \int_{\mathbb{R}} \mathbb{1}_{\{z \leq t\}} \left(\int_{\mathbb{R}} f_X(z - y) d\mathcal{P}_Y(y) \right) dz =: \int_{-\infty}^t f_Z(z) dz. \end{aligned}$$

That is, a function $f(z) : \mathbb{R} \rightarrow \mathbb{R}$ defined by the formula $\int_{\mathbb{R}} f_X(z - y) d\mathcal{P}_Y(y)$ is the probability density function $f_Z(z)$ by definition: its integral over any ray $(-\infty, t]$ gives the CDF of Z at the point t , for any real t .

Finally, since Y has density f_Y , by the change of variables theorem, we conclude that

$$f_Z(z) = \int_{\mathbb{R}} f_X(z - y) d\mathcal{P}_Y(y) = \mathbb{E}[f_X(z - Y)] = \int_{\mathbb{R}} f_X(z - y) f_Y(y) dy. \quad \square$$

Remark 6.19. If X and Y are independent random variables with laws $\mathcal{P}_X = \mu$ and $\mathcal{P}_Y = \nu$ respectively, then the measure corresponding to the law of $X + Y$ is called the *convolution* of μ and ν , and is denoted by $\mu * \nu$.